APWG eCrime eXchange (eCX) New User Start-up Guide

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This instructional document is the “new user” guide to explain our processes and how to use the eCrime eXchange efficiently.

The eCX has two access paths, each with read and write functionality for the authorized user: a GUI at https://ecrimex.net, and the API at https://ecrimex.net/api/v1/<module>.

The eCX data is divided into different data modules for different types of data, representing different kinds of cybercrime events (e.g. establishment of a phishing URL, registration of a brand-abusing domain name for a phishing campaign, etc.).

This division allows for each module to hold different *data elements* in records that represent and archive specific kinds of common cybercriminal events. For example, the phish module has a 'URL' field as that's essential for phishing collectors. The malicious domain module does not have a URL field because the data records hold DNS names.

The User Onboarding Process

The overall user experience follows this path:

1. Initial eMail Invitation. eCX users are sent an invitation by email to join the platform. Clicking on the invitation link allows the new user to setup an eCX account - entering their name, selecting a password, etc. - then clicking OK to submit the new account for validation by eCX personnel.
2. eCX User Validation. The system administrators are notified when a user sets up their initial account and approves the new user's account. There may be a delay in the user setting up the account and the system admins performing their task.
3. Initial Module Access. A new user has read access to the phish data module by default. This access is for both the GUI and the API. We suggest that a new user logs into the eCX GUI (https://ecrimex.net) to acquaint themselves. Roll-over "modules" (at the top of the page), and select "phish" to view the current data in the phish module.
4. The User Profile. At the top right of the GUI page is your username. Clicking on it will show two options: "Log Out" and "Profile". Log out will do just that. Your profile shows:
    a. Information about you;
    b. Information about your organization's membership status;
    c. Your API key. The key can be changed by clicking on the icon at the right of the key display.

5. Changing your Password. The easiest way to change your password is to visit https://ecrimex.net and click the “forgot your password?” link.
6. If your co-workers move on from your organization, we would appreciate a note letting us know so we can disable their accounts, reducing our password exposure on unused accounts.

The Modules

The ECX data modules are:

1. Phish. This data set contains URLs of phishing sites and the brand they are impersonating.
2. ReportPhishing. This contains the slightly-processed emails sent by the public to our reportphishing@apwg.org email address. Normally receives 300,000-500,000 monthly.
3. Malicious_domain. DNS domains that are a fake store front or serving malicious content.
4. Malicious_IP. IP Addresses that appear to only attack or probe others on the Internet.
5. Cryptocurrency-addresses. Records of crypto WalletIds seen in crimeware.
6. Malicious_SMS. This is data retrieved from phish or malicious SMS or TXT messages.

Requesting read access to a module

To gain access to a specific module, log into the GUI, select the module of interest, read the description, and click the “Request Access” button. System admins will change your permissions.

Requesting submission access to a module

Access to a module in the GUI gives you the ability to submit data via the GUI.

Submission via the API requires the submission privilege for that module. We ask that you test and perfect your API scripts using the eCX sandbox before asking for submission rights to the production eCX API. We have learned our lessons.

Confidence Factors

- All of the eCX API /endpoints require a record to have a **Confidence Factor** that relates to the degree of manual, human processing used in scoring the report as an indicator of the its reliability. As follows:
- **100 CF** = Manual inspection/confirmation by an APWG member reporter, submitting data under terms set out in the APWG Data Sharing Agreement
- **90 CF** = Automated inspection/discovery by Artificial Intelligence or Machine Learning systems directly curated by an APWG member submitter, under terms set out in the APWG Data Sharing Agreement. *Users are encouraged to verify records at this level before indiscriminately blocking them or using them in forensic routines*.
- **50 CF** = Machine data-capture or third-party volunteered report ***without*** confirmation by APWG member personnel or by a member-curated AI/ML
    - Third-party source or general public sourcing so - on its face, without further meta-data that would speak to reliability. The assumption is that the record has a 50/50 chance of being accurate, therefore requiring user confirmation.
- **0 CF** = False positive

Updating Confidence Factors

eCX users may use the GUI or the API to “annotate” data records to add to their confidence or mark a record as a false positive.

False positives have a confidence value of 0 (zero). Log into the eCX GUI, find the record you wish to modify, double click it, change the confidence to 0, and click update. You can also change the confidence value of a record via a PUT command in the API.

Using the API

The Sandbox.

There is a sandbox of the eCX that periodically gets data inserted for testing. It has no direct connection to the production environment. It is located at https://sandbox.ecrimex.net, and your API key should work just fine there.

Accessing eCX data via the API:

For each API request, your API key should reside in an HTTP Authorization header, like "Authentication: Bearer api-key-value"

As an example, using curl for access to phish

- curl -k https://sandbox.ecrimex.net/api/v1/phish -H "Authorization: Bearer 0123my-api-key890" -H '"Content-Type": "application/json"'

Using a python script for API access.

```
def get:
headers = {
      "Authorization": f"Bearer " + CONFIG['api_key'],
      "Content-Type": "application/json",  # Specify content type for the request
   }
    resp = requests.ge("https://sandbox.ecrimex.net/api/v1/phish", headers=headers, verify=False )
   print("API Response: " + str(resp.status_code)+ " " + str(resp.content))
```

Users may also use their own scripts in any programming language for access.

Documentation Available in the eCX GUI:

User Guide, available under the 'Docs' link.

API Swagger Documentation, available in the ''Swagger API Docs " tab.

Documentation Available in the APWG git:

There are some sample python3 scripts to access the eCX via the API at:

https://github.com/APWG/ecx3-api-routines

Many Frequently Asked Questions (FAQ) are answered at:

https://apwg.github.io/ecx2-faq/

Assistance and Support.

Help is always available via email at support@ecrimex.net. Our support staff tries to respond promptly. Please include the error message or API call you are having issues with for niceness.

Join the announcement mail list to let users know if things are broken. Sign up at: https://members.apwg.org/mailman3/lists/ecx-announce.members.apwg.org/

Check out the new eCX-users mail list, too, at: https://members.apwg.org/mailman3/lists/ecx-users.members.apwg.org/

As APWG members, you will be added to the members list which is good for general queries and questions about new attackers, techniques, or odd results.

We have a discord server, but we are not yet exactly sure what to do with it. Please send your suggestions regarding the development of this resource to: support@ecrimex.net

